K.RAMAKRISHNAN
COLLEGE OF ENGINEERING
An Autonomous Institution
Permanently Affiliated to Anna University Chennai, Approved by AICTE New Delhi,
ISO 9001:2015, 14001:2015 certified institution, Accredited by NBA and with A grade by NAAC
Samayapuram, Tiruchirappalli – 621 112, Tamilnadu, India.

A PROJECT REPORT

on

CORAL BLEACHING EARLY – WARNING BUOY

Submitted in partial fulfillment of requirements for the award of the course of

GEA1401 – ENVIRONMENTAL SCIENCE(TCPR)

Under the guidance of

Mr. R. Balamurugan, M.E., (Ph. D.,)

Assistant Professor/ECE

Submitted By

AARTHI S (8115U23EC001)
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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

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SAMAYAPURAM – 621 112

NOVEMBER 2026

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BONAFIDE CERTIFICATE

Certified that this project report titled "**CORAL BLEACHING EARLY-WARNING BUOY**" is the bonafide work of **AARTHI S (8115U23EC001), BAVADHARANI P (8115U23EC008), MADHUMITHA T (8115U23EC059), MEERA B (8115U23EC062)** who carried out the project work under my supervision. Certified further, that to the best of my knowledge the work reported here in does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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Submitted for the End Semester Examination held on …….

INTERNAL EXAMINER **EXTERNAL EXAMINER**

DECLARATION

I jointly declare that the project report on "CORAL BLEACHING EARLYWARNING BUOY" is the result of original work done by us and best of our knowledge, similar work has not been submitted to **"ANNA UNIVERSITY CHENNAI"** for the requirement of Degree of BACHELOR OF ENGINEERING. This project report is submitted on the partial fulfillment of the requirement of the award of degree of BACHELOR OF ENGINEERING.

Signature

AARTHI S
BAVADHARANI P
MADHUMITHA T
MEERA B

Place: Samayapuram
Date:

ACKNOWLEDGEMENT

It is with great pride that we express our gratitude and indebtedness to our institution, "**K. Ramakrishnan College of Engineering (Autonomous)**", for providing us with the opportunity to do this project.

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We render our sincere thanks to the Course Coordinator and other staff members for providing valuable information during the course.

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VISION

To achieve a prominent position among the top technical institutions

MISSION

- To bestow standard technical education par excellence through state-of-the-art infrastructure, competent faculty and high ethical standards.
- To nurture research and entrepreneurial skills among students in cutting edge technologies.
- To provide education for developing high-quality professionals to transform the society.

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VISION

To be distinguished as a prominent program in Electronics and Communication Engineering Studies by preparing students for Industrial Competitiveness and Societal Challenges.

MISSION

M1. To equip the students with latest technical, analytical and practical knowledge

M2. To provide vibrant academic environment and Innovative Research culture

M3. To provide opportunities for students to get Industrial Skills and Internships to meet out the challenges of the society.

PROGRAM EDUCATIONAL OBJECTIVES (PEO'S)

PEO1: Graduates will be competent enough to design, develop and integrate complex industrial products for modern electronics and communication systems.

PEO2: Graduates will be able to propose research oriented, industry relevant design solutions in state-of-the-art electronics and communication domains.

PEO3: Graduates will be able to engage in start-up and entrepreneurial activities developing innovative prototypes for the challenging needs of the society.

PROGRAM SPECIFIC OUTPUTS (PSO'S):

PSO1: Analyze and design complex electronic circuits, communication and embedded systems by applying the principles of engineering and mathematics.

PSO2: Analyze and provide innovative design solutions to research problems, industrial problems and for the demanding needs of the society in the domains of electronic integrated circuits, communication systems and embedded VLSI.

PSO3: Continuously involve in life-long learning activities so as to evolve and become an active team member in diverse and challenging design situations.

PROGRAM OUTCOMES

PO1: Engineering knowledge: Apply the knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to the solution of complex engineering problems.

PO2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development (WK1 to WK4).

PO3: **Design/Development of Solutions:** Design creative solutions for complex engineering problems and design /develop systems/ components/ processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)

PO4: **Conduct Investigations of Complex Problems:** Conduct investigations of complex engineering problems using research-based knowledge including

design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

PO6 The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).

P07: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)

PO8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences

PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

SUSTAINABLE DEVELOPMENT GOALS (SGDs)

☐ **Goal 1: No Poverty** – End poverty in all forms everywhere

☐ **Goal 2: Zero Hunger** – End hunger and improve nutrition

☐ **Goal 3: Good Health and Well-being** – Ensure healthy lives

☐ **Goal 4: Quality Education** – Provide inclusive, equitable education

☐ **Goal 5: Gender Equality** – Empower women and girls

☐ **Goal 6: Clean Water and Sanitation** – Ensure safe water and sanitation

☐ **Goal 7: Affordable and Clean Energy** – Access to sustainable energy

☐ **Goal 8: Decent Work and Economic Growth** – Promote jobs and growth

☐ **Goal 9: Industry, Innovation and Infrastructure** – Build resilient infrastructure

☐ **Goal 10: Reduced Inequalities** – Reduce inequality

☐ **Goal 11: Sustainable Cities and Communities** – Make cities safe and sustainable

☐ **Goal 12: Responsible Consumption and Production** – Reduce waste

☑ **Goal 13: Climate Action** – Combat climate change

☑ **Goal 14: Life Below Water** – Protect oceans

☐ **Goal 15: Life on Land** – Protect ecosystems and biodiversity

☐ **Goal 16: Peace, Justice and Strong Institutions** – Promote justice and peace

☐ **Goal 17: Partnerships for the Goals** – Strengthen global partnerships

CO – PO MAPPING FOR PROJECT WORK

After successful completion of this course, the students should be able to

CO1: Understand environmental concepts, structure, function and importance of ecosystems and biodiversity in maintaining ecological balance.

CO2: Assess sustainable methods for the management of natural resources to address current and future demands.

CO3: Identify sources, impacts, and control measures of various types of pollution and solid waste.

CO4: Interpret and apply relevant environmental policies and legislation for resource and environmental protection.

CO5: Develop innovative solutions, reports, models and presentations to address sustainable practices.

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	✓	✓	✓	✓	-	-	-	-	-	-	-	-	✓	-
CO2	✓	✓	✓	✓	-	-	-	-	-	-	-	-	✓	-
CO3	✓	✓	✓	✓	-	-	-	-	-	-	-	-	✓	-
CO4	✓	✓	✓	✓	-	-	-	-	-	-	-	-	✓	-
CO5	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

ABSTRACT

The “Coral Bleaching Early-Warning Buoy” is an innovative, solar-powered embedded IoT solution designed to provide continuous, real-time subsurface monitoring of vulnerable marine ecosystems to prevent climate-induced reef degradation. While conventional monitoring relies on wide-area satellite data or delayed manual surveys that fail to catch localized stress, this low-cost system uses an ESP32 microcontroller interfaced with a waterproof DS18B20 temperature sensor and a Light Dependent Resistor (LDR) to track environmental stress directly at the reef site. The system continuously processes these live data streams against scientifically established coral thermal-tolerance limits. Under normal conditions, a green LED indicates a safe baseline; however, if thermal thresholds are crossed, the microcontroller instantly triggers local audible and visual alarms using a buzzer and red LED, while an integrated Wi-Fi module transmits remote alerts to a mobile app or dashboard. Powered autonomously by an integrated solar matrix and a rechargeable battery backup, this scalable framework enables proactive, early-stage conservation interventions, aligning directly with UN Sustainable Development Goals 13 (Climate Action) and 14 (Life Below Water).

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CHAPTER 1
INTRODUCTION

1.1 INTRODUCTION

The “Coral Bleaching Early-Warning Buoy” is an innovative ecological solution designed to provide reliable and continuous marine environmental monitoring even in remote, infrastructure-lacking ocean zones. It addresses the major limitations of conventional reef tracking methods, which rely on widearea satellite predictions or delayed manual diving surveys that fail to detect stress before irreversible damage occurs. Equipped with an underwater temperature sensor and an encapsulated light sensor linked to a microcontroller for precise data processing, the system detects micro-climatic shifts and triggers instant alerts through a buzzer sound and LED indicators. A sustainable power chain ensures a stable regulated output from a rechargeable battery backed by a solar panel, maintaining uninterrupted operation and autonomous field deployment. Compact, affordable, and energy-efficient, this system enhances marine ecosystem preparedness and is suitable for vulnerable reefs, marine sanctuaries, and coastal conservation zones globally.

1.2 OBJECTIVES

The Coral Bleaching Early-Warning Buoy offers the following key characteristics:

- Work Without Grid Power: Use an integrated solar panel and rechargeable battery to keep running autonomously in remote ocean zones.
- Give Multi-Mode Alerts: Send immediate danger warnings through a local buzzer, high-visibility LEDs, and remote IoT dashboard messages.
- Save Energy: Maintain continuous 24/7 environmental monitoring with highly optimized, low-power embedded components.
- Easy to Deploy: Simple, cost-efficient architecture suitable for widespread installations by marine researchers and conservation agencies.
- Fast Response : React instantly when underwater temperature or light levels cross scientifically established safe thresholds.
- Protect Life Below Water: Help reduce coral mortality and safeguard marine

biodiversity by providing an early window for human intervention.

- Detect Stress Factors: Sense marine thermal changes and light intensity changes quickly using underwater DS18B20 and LDR sensors.

1.3 FEATURES

1. Marine Parameter Sensing Unit: Detects underwater temperature and ambient light intensity in real time using waterproof sensors for early bleaching

2. Microcontroller Control: Processes environmental data and triggers threat alerts automatically when thermal or light values exceed a set safe limit.

3. Solar Harvest Power Supply: Provides a sustainable, self-sufficient energy system using an integrated solar panel to charge the internal storage bank.

4. Uninterrupted Operation: Keeps the continuous monitoring loop active 24/7 across remote reef locations with a reliable rechargeable battery backup.

5. Alerts: Local high-decibel buzzer, dual-state LEDs, and optional remote WiFi dashboard telemetry.

CHAPTER 2
EXISTING SYSTEM

2.1 INTRODUCTION

Despite significant scientific advancements, reliable marine ecosystem prediction and the development of highly localized Coral Bleaching Early Warning Systems remain elusive global goals. Current conservation efforts struggle with finding reliable diagnostic indicators directly at the reef level across different global marine settings, leading to delayed responses or missing localized heat stress events entirely. Existing large-scale warning strategies rely on tracking wide-area Sea Surface Temperature (SST) anomalies via satellite remote sensing, which typically issue generalized regional alerts rather than capture micro-climatic shifts at specific depths. This creates a challenging tracking gap or "blind zone" right at the subsurface reef locations where the corals reside. A crucial need exists to develop robust, cost-effective in-situ electronic instrumentation, like the proposed sensor-integrated early-warning buoy, to improve local monitoring accuracy, mitigate ecological risk, and reduce the uncertain nature of localized coral mortality..

2.2 DRAWBACKS

1. Prevention Remains a Vision: Accurate prediction and site-specific tracking of incipient coral stress "still remains a vision of the future," indicating a fundamental lack of localized monitoring reliability.
2. Absence of Local Precursors: Early warning analysis suffers due to the "absence of reliable diagnostic precursors" captured directly at the subsurface depth across different marine settings.
3. High False Alarm Rate: Existing macro-scale warming monitoring programs based on satellite data have led to "many false alarms" or broad regional generalizations, risking "erroneous detections" at specific reefs.
4. Delayed Observation Window: Traditional survey methods often catch

structural degradation after a "smaller response time" for active prevention.

5. The Resolution Blind Zone: Wide-area satellite monitoring systems are ineffective in capturing the precise micro-climates within small reef zones, creating a thermal "blind zone" where there is insufficient data to execute a localized early warning.

2.3 BLOCK DIAGRAM

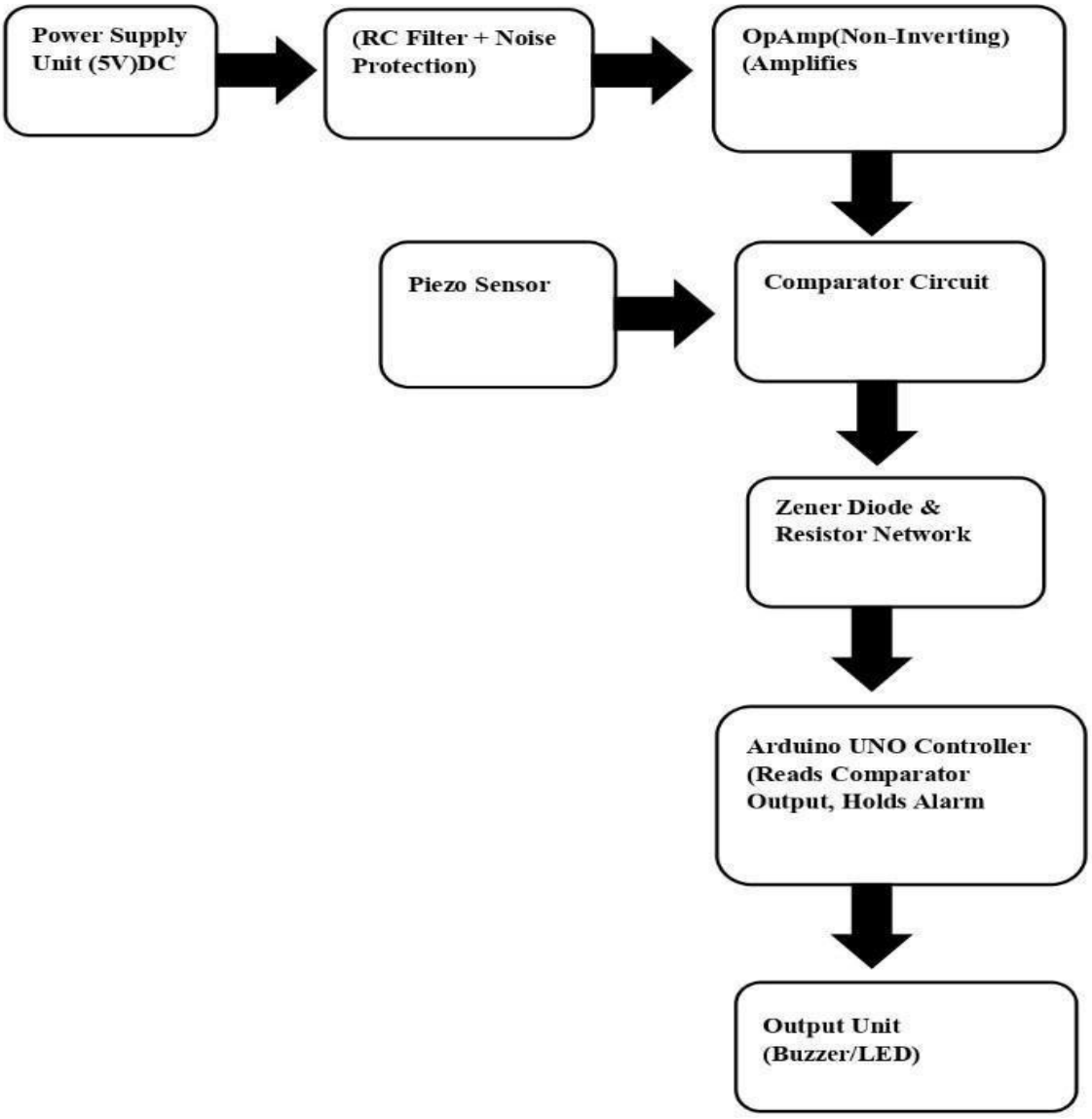


Fig 2.1 EXISTING SYSTEM BLOCK DIAGRAM

2.4 EXPLANATION

1. Sustainable Power Supply Chain :
    - Solar Panel Matrix: This acts as the primary ambient energy harvester for remote marine deployment. It converts natural sunlight into electrical energy to ensure the buoy operates sustainably in deep water without manual power maintenance.
    - TP4056 Charger Module: The voltage generated by the solar panel feeds directly into this charging manager. It regulates the power flow and provides safety cuts against overcharging to protect the life of the storage batteries.
    - 18650 Li-ion Battery Bank: This is the main energy storage core of the system. During lab prototype testing, it acts as the immediate active power source, and during field operation, it provides the 24/7 energy backup needed to keep the buoy running through the night.
    - LM2596 Buck Converter: Since raw battery voltage fluctuates, this high efficiency step-down regulator takes the battery output and smooths it into a perfectly stable 5V system power rail. This clean 5V line safely drives the delicate computing chips without risking voltage spikes.

2. In-Situ Marine Sensing Unit (System Inputs)
    - DS18B20 Waterproof Temperature Sensor: This ruggedized digital probe is submerged directly at the reef level. It continuously samples the subsurface ocean water temperature in real time and sends precision digital data frames directly to the microcontroller.
    - LDR Light Sensor: Because high solar radiation causes severe coral stress alongside high heat, this photo-sensitive sensor tracks changes in ambient light intensity over the reef and routes its readings as an analog voltage line.

3. Central Control & Multi-Mode Alert System (System Outputs) :

- Node MCU ESP Microcontroller: This serves as the master edge-computing "brain" of the entire buoy. It takes real-time inputs from both the temperature probe and the light sensor, processes the values, and matches them against preprogrammed safe biological thresholds. Based on this analysis, the controller branches into two distinct logic routes:
- Normal Status State (Green LED): If the reef parameters are inside the safe baseline limits, the controller sends power only to the Green LED, giving visual confirmation that the ecosystem is healthy.
- Bleaching Risk Alert (Red LED & Buzzer): The exact moment the water temperature or light levels cross the safe threshold, the controller cuts off the green indicator. It immediately shifts power to fire the high-visibility Red LED and sounds the high-decibel 5V Piezo Alarm Buzzer to give an instant on-site warning.
- Remote Web Dashboard: Simultaneously while triggering local alarms, the NodeMCU uses its integrated Wi-Fi radio to broadcast live telemetry over the network. This updates an online Remote Web Dashboard Interface, allowing researchers anywhere in the world to view real-time data curves and get flashing digital hazard warnings instantly.

CHAPTER 3

PROPOSED SYSTEM AND NOVELTY

3.1 INTRODUCTION

The “Coral Bleaching Early-Warning Buoy” is an IoT-based system developed to provide early alerts of marine thermal and solar stress using a waterproof DS18B20 temperature sensor and an encapsulated LDR. It is powered by an 18650 Li-ion battery regulated through an LM2596 buck converter and a TP4056 charging module, backed by a solar panel to ensure continuous power delivery during field deployment. The Node MCU microcontroller processes real-time data and compares it with predefined safe thresholds. When excessive heat or light is detected, the system triggers local alerts via a red LED and buzzer, while a green LED indicates normal reef conditions. Concurrently, the microcontroller transmits live telemetry data to a remote cloud dashboard for researcher tracking. Its low power consumption and cost-efficient design make it highly suitable for scalable marine conservation and proactive reef protection in vulnerable coastal ecosystems.

3.2 CIRCUIT DIAGRAM

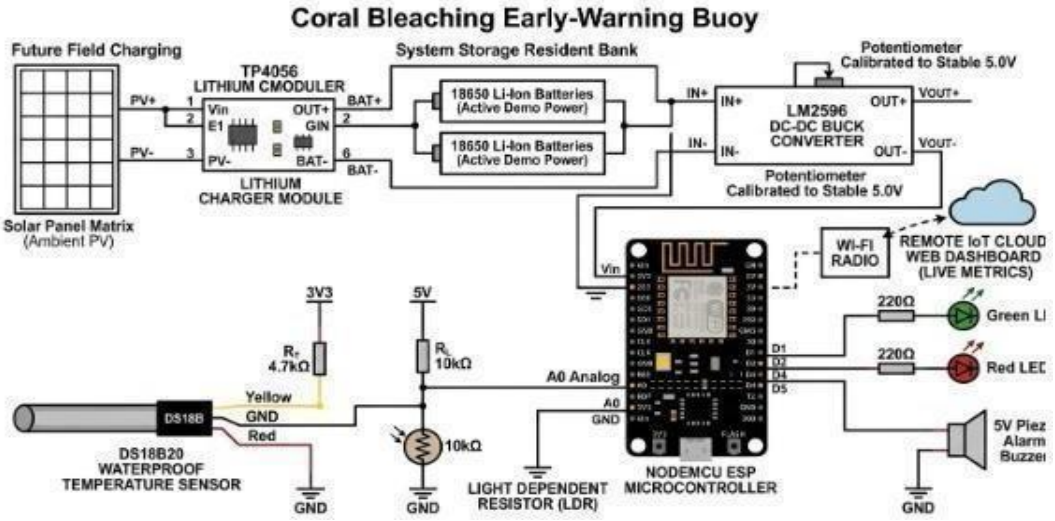


Fig 3.1 CIRCUIT DIAGRAM

3.3 HARDWARE REQUIRMENTS:

1. Solar Panel Matrix The system incorporates a Solar Panel Matrix that captures ambient sunlight and converts it into electrical energy. It serves as the primary sustainable power source intended for long-term, maintenance-free operation during real-time marine field deployment.

2. TP4056 Battery Charger Module The electrical current generated by the solar panel flows into this power conditioning module. It safely regulates the incoming voltage to charge the system storage and features integrated overcharge protection to maximize cell durability and lifespan.

3. 18650 Li-ion Battery Bank This serves as the primary energy reservoir of the entire architecture. During the prototype laboratory demo, the entire hardware setup is driven exclusively by this high-capacity lithium storage cell, ensuring a reliable power baseline that can sustain continuous 24/7 logging.

4. LM2596 Buck Converter This module serves as a highly efficient DC-DC step-down voltage regulator within the power supply network. It takes the variable fluctuating output from the lithium battery bank and steps it down into a highly stable, regulated 5V DC power rail to safely energize the downstream microcontroller and sensor logic arrays.

5. DS18B20 Waterproof Temperature Sensor This ruggedized, stainless-steel probe is the primary underwater input component responsible for tracking the real-time thermodynamic changes of the reef water. It utilizes a single-bus digital wire architecture to forward precise subsurface temperature metrics to the central controller at regular intervals.

6. Light Dependent Resistor (LDR Sensor) Excessive solar radiation and extreme light exposure are major concurrent drivers of coral reef stress. Enclosed within a transparent protective layer, this photosensitive variable resistor reads localized ambient illumination levels and converts light field fluctuations into an analog data stream.

7. Node MCU ESP Microcontroller The Node MCU acts as the definitive operational "brain" of the early-warning buoy system. It continuously reads the input metrics from the DS18B20 digital thermal probe and the LDR network, running edge-computing code that compares live data fields against scientifically established coral bleaching tolerance thresholds.

8. Output Unit (Buzzer, LEDs, and IoT Dashboard) Finally, the Output Unit handles the multi-mode threat notification cycle. Under safe baseline operations, it maintains a steady visual verification via a Green LED; upon crossing a danger threshold, it instantly sets the Green line low while triggering a Red LED, a high-decibel local Buzzer, and an automated Wi-Fi telemetry packet stream to the remote IoT Dashboard.

3.4 COST ESTIMATION

The primary costs involved in the "Coral Bleaching Early-Warning Buoy" include the power chain (solar panel, TP4056 charger, buck converter, battery), sensing (DS18B20, LDR), control (NodeMCU ESP), alerting (buzzer, LEDs, remote dashboard interface), and wiring/assembly. Choosing high-quality waterproof sensors and reliable storage cells raises upfront cost but improves durability and reduces maintenance/replacement over time in harsh marine environments.

Total Estimated Cost: ₹1,680

Itemised estimate:

Solar Panel Matrix Array (Ambient PV Energy Input): ₹250

TP4056 Lithium Battery Charger Board (Protection Node): ₹50

18650 Lithium-Ion Battery Matrix (Rechargeable Cells / Set of 2): ₹500

LM2596 Buck Converter (DC–DC step-down module): ₹150

Node MCU ESP Microcontroller Board (Wi-Fi Central Core): ₹300

DS18B20 Ruggedized Waterproof Temperature Probe: ₹200

Encapsulated Light Dependent Resistor (LDR Sensor): ₹30

Buzzer & LED Indicator Array (Red LED, Green LED, 5V Piezo): ₹50

General Purpose Dot PCB Board: ₹150

Connecting Wires & Sealing Assembly Material: ₹150

3.5 WORK FLOW DIAGRAM:

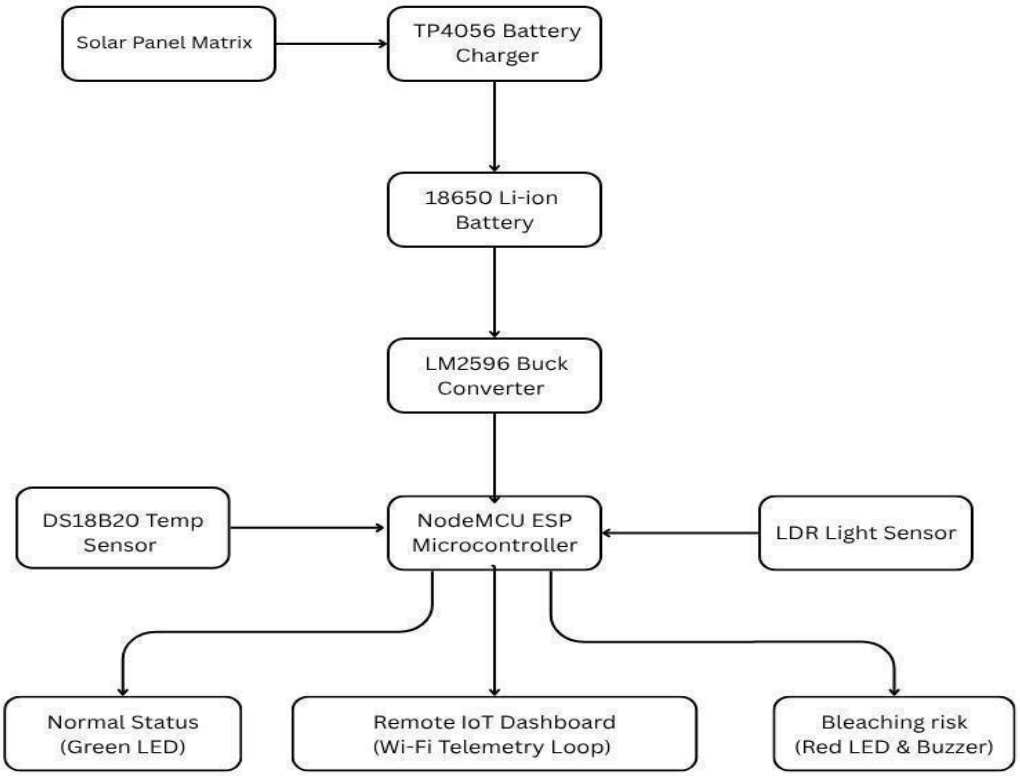


Fig 3.2 PROPOSED SYSTEM BLOCK DIAGRAM

The proposed workflow diagram illustrates an autonomous environmental safety system driven by a horizontal power chain and a vertical sensing matrix. The power network begins at the Solar Panel Matrix, which feeds through a Lithium Charger Module to safely replenish a Rechargeable Battery (18650), providing stored voltage to a Buck Converter (DC-DC) to deliver a clean power rail for downstream electronics. Concurrently, the sensory layer uses a Temperature Sensor (DS18B20) and a Light Sensor (LDR) to capture subsurface marine variables, streaming these real-time input metrics straight into the Microcontroller (Node MCU ESP). The controller processes these data streams against pre-set biological safety thresholds to manage system outputs; under safe conditions, it maintains a normal green indicator, but the moment a threshold is

crossed, it updates an online Remote Web Dashboard while simultaneously triggering local visual and audio alarms via the LED Alerts (Green/Red) and a Local Alert Buzzer .

3.6 NOVELTY

The Coral Bleaching Early-Warning Buoy offers the following key characteristics:

- Dual-Alert System: The system gives warnings through a local buzzer, flashing LEDs, and a remote web dashboard so conservationists get notified even if one method fails.
- Instant Subsurface Sensing: Using the waterproof DS18B20 sensor and Node MCU ESP, it detects marine thermal shifts instantly and reacts quickly to even small temperature variations.
- Autonomous Power Chain: A buck converter with solar-backed battery storage keeps the system working 24/7, even in remote ocean zones without grid infrastructure.
- Low-Impact Design: Uses low-power components, making the device small, energy-efficient, and easy to deploy directly over vulnerable reef locations.
- Future-Proof Upgrades: The design can be upgraded later with AI models, cloud historical logging, or machine learning for better coral mortality prediction and wide-area trend analysis.
- Proactive Protection: Alerts researchers globally and field teams nearby, improving overall reef safety and allowing for quick preventive interventions before whitening occurs.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 RESULT

This chapter presents the experimental results and performance analysis of the "Coral Bleaching Early-Warning Buoy." The developed system is designed to detect subsurface environmental shifts in real time and provide instant multi-mode alerts using a buzzer, LEDs, and a remote cloud dashboard. It combines the sensing capability of the DS18B20 waterproof temperature sensor and an LDR light sensor with the processing power of the Node MCU ESP microcontroller, ensuring fast and accurate coral bleaching risk detection. A stable and uninterrupted power supply is achieved through a buck converter with a solar-backed battery storage matrix, allowing the system to remain functional even in remote ocean environments—an essential feature for continuous marine conservation scenarios. Key performance parameters such as sensor response time, threshold sensitivity, and wireless telemetry reliability are evaluated to verify system effectiveness.

4.2 OUTPUT



Fig 4.1 REAL TIME OUTPUT

4.3FEEDBACK ANALYSIS

1. Customer 1 (Marine Biologist / Field Researcher)

   Suggestion: "A high-visibility strobe light or heavier auxiliary horn should be added to warn passing offshore vessels in heavy weather."

   This feedback highlights the need for stronger and more visible local alert mechanisms, especially in low-visibility or stormy marine environments where a simple low-power buzzer and standard LED may not be noticed by nearby boats.

   Future enhancements will include testing different types of ruggedized marine alarms, optimizing their visibility range across open water, and adding adjustable alert modes to suit various coastal environments such as shallow reefs, lagoons, and public marine sanctuaries.

2. Customer 2 (Environmental Conservationist):

   Suggestion: "An automated cloud logging framework and long-term remote data interface must be included to track global trends without

needing physical maintenance access to the buoy."

We improved the design by adding an integrated Wi-Fi communications subsystem that streams wireless TCP/IP telemetry packets continuously to a live online Remote IoT Web Dashboard.

Further evaluation will be done to test cloud synchronization efficiency, data transmission stability, and power regulation consumption rates under constant long distance monitoring conditions. This suggestion reflects users' concern that the system must remain globally accessible at all times, ensuring that critical pre-bleaching alerts and historical thermal curves are not lost during remote deployment.

4.3.1 APPLICATION OF THE PRODUCT

- Marine Sanctuaries & Reef Zones: Ideal for deployment over vulnerable coral colonies to track thermal stress and prevent sudden mortality.
- Academic Marine Research Platforms: Useful for environmental science institutions to study subsurface temperature variations and solar radiation impacts.
- Coastal Management Networks: Beneficial for conservation agencies to build low-cost baseline data grids for widespread aquatic resource protection.
- Remote Offshore Monitoring Areas: Suitable for isolated maritime zones lacking grid infrastructure due to solar harvesting and battery power.
- Commercial Aquaculture Facilities: Adaptable for open-water hatcheries and fish farms to monitor local temperature variables and safeguard stock.

4.3.2 FEEDBACK QUESTIONNAIRES AND ANALYSIS

a. **Improve Bleaching Threshold Accuracy:** Users suggested refining the edge detection algorithm to reduce false alerts caused by short-term daily temperature spikes or temporary cloud shadows passing over the sensor grid.

b. **Dashboard Interface Enhancement:** Several users recommended adding more analytical details to the live Remote Web Dashboard, such as maximum daily thermal stress accumulation indices, historical baseline averages, and precise battery voltage tracking for better telemetry management.

c. Power Harvesting Resilience: Feedback indicated the need for integrating a more sensitive or larger surface solar matrix panel to ensure fast cell replenishment during periods of heavy overcast weather or monsoon cloud cover.

CHAPTER 5

MARKET OPPORTUNITITES

5.1 SOCIAL BENEFITS:

- Empowers Coastal Communities: Protects the health of coral reefs, which directly sustains local tourism and small-scale artisanal fishing economies that coastal populations rely on for their daily livelihoods.
- Secures Natural Storm Barriers: Protects the physical structure of local reefs, preserving their natural ability to absorb up to 97% of wave energy and preventing catastrophic coastal erosion and storm surge damage to seaside towns.
- Raises Public Conservation Awareness: Delivers transparent, real-time local environmental metrics that can be integrated into civic schools and public education programs to foster environmental responsibility.
- Supports Global Ecological Sustainability: Provides open-source, affordable monitoring data that aids underfunded coastal regions, aligning directly with UN Sustainable Development Goals 13 (Climate Action) and 14 (Life Below Water).

5.2 ECONOMIC BENEFITS:

- Saves Local Tourism Income: Safeguards the vibrant health of coral reefs, which directly protects billions of dollars in global ecotourism revenue, dive sports, and coastal hospitality jobs [Vedantu NCERT EVS].
- Lowers Natural Disaster Damages: Preserves the structural barrier of healthy reefs that naturally break heavy wave energy, saving coastal cities massive infrastructure repair expenses from storm surges and erosion.

- Reduces Conservation Technology Costs: Replaces expensive, highend commercial maritime instruments with an affordable, low-cost opensource IoT architecture, making widespread ecological monitoring financially feasible for developing nations.

5.3 TECHNOLOGICALADVANCEMENTS:

1. Satellite Data Cross-Referencing: Integrates local subsurface sensor telemetry with wide-area NOAA Coral Reef Watch satellite imagery to create a highly accurate hybrid environmental model.
2. LoRa WAN Mesh Network Expansion: Upgrades the communication protocol to long-range wireless networks, allowing hundreds of offshore buoys to transmit data effortlessly over 15+ kilometers without relying on cellular data.
3. Edge Machine Learning Models: Implements lightweight predictive AI algorithms directly on the ESP microcontroller to calculate future bleaching indices based on real-time water temperature and light accumulation curve

CHAPTER 6

CONCLUSION AND FUTURE SCOPES

6.1 CONCLUSION

The Coral Bleaching Early-Warning Buoy provides real-time monitoring of underwater temperature and light intensity using a waterproof DS18B20 temperature sensor, an LDR sensor, and an ESP8266/NodeMCU microcontroller. It generates immediate alerts through a buzzer and LED indicators when environmental parameters cross predefined safety thresholds. The Wi-Fi module enables remote monitoring through an IoT dashboard, while the rechargeable battery and solar power system support continuous operation.

This project supports **SDG 14 – Life Below Water** by helping protect coral reefs and marine biodiversity through continuous environmental monitoring and early-warning alerts. It contributes to **SDG 13 – Climate Action** by monitoring temperature changes associated with climate-related stress on coral reefs and supporting timely conservation measures. It also promotes sustainable technology through an affordable, energy-efficient, IoT-based marine monitoring system.

6.2 FUTURE SCOPES

1.AI-Based Bleaching Prediction

Use machine learning to analyse temperature and light data and predict coral bleaching risks before severe damage occurs.

2. Underwater Sensor Network Expansion

Deploy multiple buoys at different depths and locations across coral reefs.

3. Automatic Marine Conservation Alerts

Integrate the buoy with a mobile application to send instant alerts to marine researchers and conservation teams

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